STPM 2024 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

Section A [15 marks]

1. If P is the pressure, A is the area and Q is the rate of change of mass, the expression of $\frac{PA}{Q}$ has the same dimension as

A force B energy C velocity D acceleration

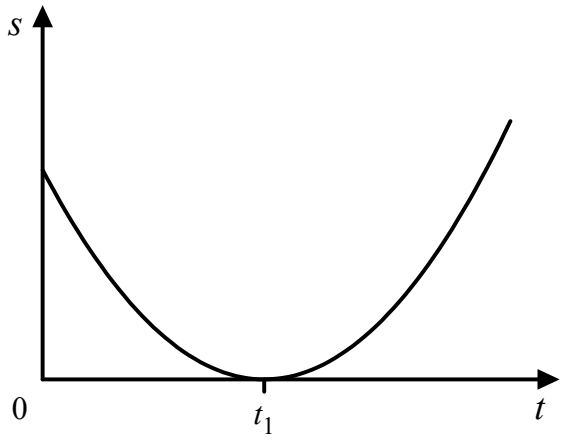
Answer: C

$$[PA] = \text{N} = \text{kg m s}^{-2};\ [Q] = \text{kg s}^{-1}$$

$$[\frac{PA}{Q}] = \frac{\text{kg m s}^{-2}}{\text{kg s}^{-1}} = \text{m s}^{-1} \rightarrow \text{velocity}$$

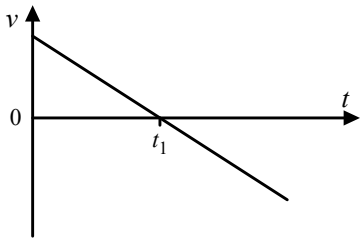
Explanation: Pressure × area is a force. Dividing a force by a rate of change of mass (kg s^{-1}) leaves m s^{-1}, the unit (and dimension LT^{-1}) of velocity. This is the idea behind rocket thrust: force = (rate of change of mass) × velocity.

2. The variation of displacement, s, with time, t, for a body is shown in the graph below.

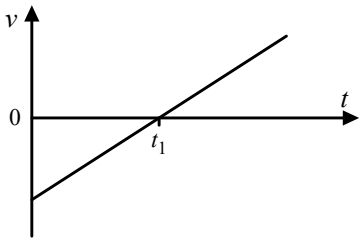


Which graph shows the correct variation of velocity, v, with time, t, of the body?

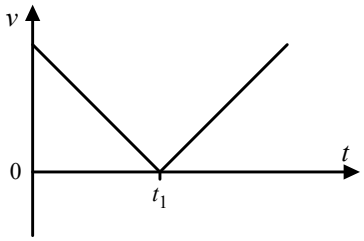
A



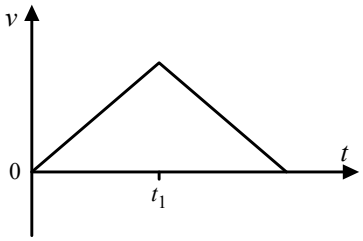
B



C



D



Answer: B

The s–t graph is a parabola with its minimum at t_1, $s \propto (t - t_1)^2$, so $v = \frac{ds}{dt} \propto (t - t_1)$.

The gradient is negative before t_1, zero at t_1 and positive after: a straight line rising through zero at t_1.

Explanation: Read the gradient of the displacement–time graph at several points: steep and negative at the start, flatter and still negative, zero at the lowest point, then positive and increasing. The velocity changes uniformly from negative to positive, so the v–t graph is a straight line crossing the time axis at t_1.

3. A particle hits a wall perpendicularly with speed u. Which statement is true about the elastic and inelastic collisions between the particle and the wall?

	Elastic collision	*Inelastic collision*
A	The particle rebounds with speed u. Momentum and kinetic energy are conserved.	The particle does not rebound. Momentum and kinetic energy are conserved.
B	The particle rebounds with speed less than u. Momentum and kinetic energy are conserved.	The particle rebounds with speed less than u. Momentum is conserved but kinetic energy is not conserved.
C	The particle rebounds with speed less than u. Momentum and kinetic energy are conserved.	The particle does not rebound. Momentum and kinetic energy are conserved.
D	The particle rebounds with speed u. Momentum and kinetic energy are conserved.	The particle rebounds with speed less than u. Momentum is conserved but kinetic energy is not conserved.

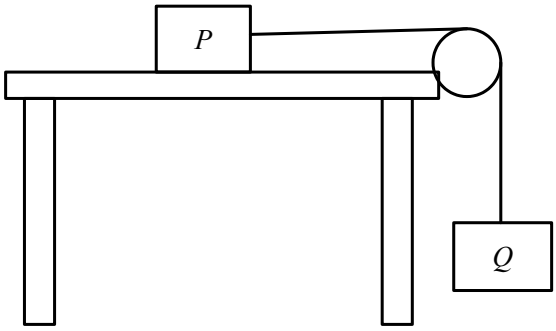
Answer: D

Elastic collision: no loss of kinetic energy, so the particle rebounds with the same speed u.

Inelastic collision: some kinetic energy is lost, so the particle rebounds with a smaller speed; momentum (of the particle and wall/Earth system) is still conserved.

Explanation: Momentum is conserved in every collision when no external force acts on the system; what distinguishes elastic from inelastic collisions is kinetic energy. Options A and C wrongly say kinetic energy is conserved in an inelastic collision.

4. A block, P, of mass 2.0 kg on a table is connected to a block, Q, of mass 4.0 kg by a light string that passes over a frictionless pulley as shown in the diagram below.



If the system is released from rest and moves with an acceleration of 2.0 m s^{-2}, what is the coefficient of kinetic friction between P and the surface of the table?

A 0.46 B 0.67 C 1.4 D 1.8

Answer: C

$$Q: m_Q g - T = m_Q a; \quad P: T - f = m_P a$$

$$m_Q g - f = (m_P + m_Q)a \Longrightarrow (4.0)(9.81) - f = (6.0)(2.0) \Longrightarrow f = 27.2 \text{ N}$$

$$\mu_k = \frac{f}{m_P g} = \frac{27.2}{(2.0)(9.81)} = 1.4$$

Explanation: Treat the two blocks as one system moving with the same acceleration: the weight of Q drives the system and friction on P opposes it. The normal reaction on P is only the weight of P, so $f = \mu m_P g$.

5. There are two machines of the same type, X and Y. If the efficiencies of X and Y are 20% and 60% respectively, then

A Y has higher rate of heat dissipation than X
B Y has higher output power than X
C Y has lower output power than X
D Y does more work than X

Answer: B

$$\text{Efficiency} = \frac{\text{output power}}{\text{input power}} \times 100\%$$

For machines of the same type (same input power), the machine with the higher efficiency gives the higher output power, so Y has a higher output power than X.

Explanation: X wastes 80% of its input as heat while Y wastes only 40%, so X has the higher rate of heat dissipation (A is wrong). Efficiency compares rates, so it does not say which machine does more work in total, because that depends on how long each machine runs (D is wrong).

6. A stone of mass 2.00 kg is tied at one end of a string of length 0.500 m and whirled in a horizontal circle. The maximum tension of the string is 36.0 N. Which minimum speed of the stone that would break the string corresponds correctly to the motion of the stone immediately after the string is broken?

	Minimum speed of the stone	*Motion of the stone*
A	3.00 m s^{-1}	In the tangential direction from the circle
B	9.00 m s^{-1}	In the tangential direction from the circle
C	3.00 m s^{-1}	In the circular direction with the same radius
D	9.00 m s^{-1}	In the circular direction with the same radius

Answer: A

$$T = \frac{mv^2}{r} \Longrightarrow v = \sqrt{\left(\frac{Tr}{m}\right)} = \sqrt{\left(\frac{(36.0)(0.500)}{2.00}\right)} = 3.00 \text{ m s}^{-1}$$

When the string breaks there is no centripetal force, so the stone moves off along the tangent.

Explanation: The tension provides the centripetal force; the string breaks when the required centripetal force exceeds the maximum tension. Once the string breaks, by Newton's first law the stone continues in a straight line along the tangent at the point where it was released (ignoring gravity, it does not keep moving in a circle).

7. A planet, P, has mass M, radius R and gravitational acceleration of free fall g. If another planet, Q, has mass $30M$ and radius $4R$, what is the gravitational acceleration of free fall for Q in terms of g?

A $56g$ B $7.5g$ C $1.9g$ D $0.5g$

Answer: C

$$g = \frac{GM}{R^2} \Longrightarrow \frac{g_Q}{g} = \frac{30}{4^2} = 1.875$$

$$g_Q = 1.9g$$

Explanation: The gravitational field strength at the surface is proportional to the mass and inversely proportional to the square of the radius. Remember to square the radius ratio: $4^2 = 16$, not 4 (which would give option B).

8. A satellite of mass 2.5×10^3 kg orbits the Earth at 1.5×10^7 m above the surface of the Earth. What is the gravitational force that acts on the satellite?

A 2.5×10^4 N B 1.3×10^4 N C 4.4×10^3 N D 2.2×10^3 N

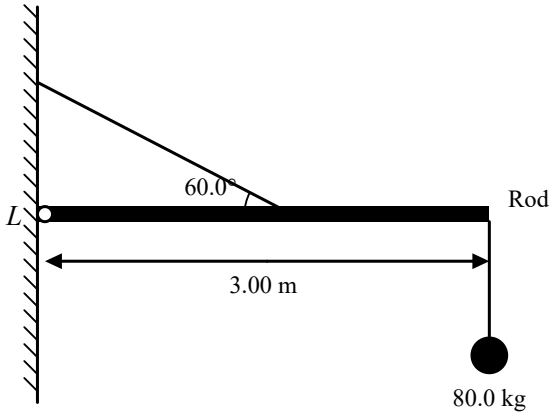
Answer: D

$$r = R + h = 6.38 \times 10^6 + 1.5 \times 10^7 = 2.14 \times 10^7 \text{ m}$$

$$F = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(2.5 \times 10^3)}{(2.14 \times 10^7)^2} = 2.2 \times 10^3 \text{ N}$$

Explanation: The distance in Newton's law of gravitation is measured from the centre of the Earth, so add the radius of the Earth to the height of the orbit. Using only the height (1.5×10^7 m) gives option C.

9. A uniform rod of mass 25.0 kg and length 3.00 m is hinged at a point, L, as shown in the diagram below.



A strong cable is attached at 2.50 m from L at an angle of 60.0° from the horizontal. If an object of mass 80.0 kg is hung at the end of the rod, what is the tension in the cable?

A 1.26 kN B 2.18 kN C 2.35 kN D 2.72 kN

Answer: A

Take moments about the hinge L:

$$(T \sin 60.0°)(2.50) = (25.0)(9.81)(1.50) + (80.0)(9.81)(3.00)$$

$$T = \frac{2722}{2.50 \sin 60.0°} = 1.26 \times 10^3 \text{ N} = 1.26 \text{ kN}$$

Explanation: Taking moments about the hinge removes the unknown hinge force. Only the vertical component of the tension ($T \sin 60.0°$) has a moment about L. The weight of the uniform rod acts at its centre, 1.50 m from L.

10. A rod of diameter 30.0 mm is pulled by a force of 300 N. What is the stress of the rod?

A 1.33×10^6 N m^{-2} B 4.24×10^5 N m^{-2} C 3.33×10^5 N m^{-2} D 1.06×10^5 N m^{-2}

Answer: B

$$A = \pi(0.0150)^2 = 7.07 \times 10^{-4} \text{ m}^2$$

$$\sigma = \frac{F}{A} = \frac{300}{7.07 \times 10^{-4}} = 4.24 \times 10^5 \text{ N m}^{-2}$$

Explanation: Use the radius (half of 30.0 mm = 0.0150 m) to find the cross-sectional area. Using the diameter as the radius gives option D (1.06×10^5).

11. Oxygen gas of mass 10 g is contained in a flask of volume 5.0×10^{-4} m^3 at 300 K. After being contained in the flask for some time, the remaining mass of the gas reduces and the pressure of the gas is 1.3×10^5 Pa. What is the mass of the gas that was used?
[Molar mass of oxygen is 32 g.]

A 9.9 g B 9.2 g C 0.83 g D 0.081 g

Answer: B

$$\text{Remaining: } n = \frac{pV}{RT} = \frac{(1.3 \times 10^5)(5.0 \times 10^{-4})}{(8.31)(300)} = 0.0261 \text{ mol}$$

$$\text{Remaining mass} = (0.0261)(32) = 0.83 \text{ g}$$

$$\text{Mass used} = 10 - 0.83 = 9.2 \text{ g}$$

Explanation: The ideal gas equation gives the number of moles still in the flask; multiply by the molar mass for the mass. The question asks for the mass *used*, so subtract from 10 g: 0.83 g (option C) is the mass that remains.

12. The kinetic energy of a diatomic gas molecule at room temperature, T, is

A $\frac{1}{2}kT$ B $\frac{3}{2}kT$ C $\frac{5}{2}kT$ D $\frac{7}{2}kT$

Answer: C

Diatomic molecule: 3 translational + 2 rotational = 5 degrees of freedom; $E = 5 \times \frac{1}{2}kT = \frac{5}{2}kT$

Explanation: By the principle of equipartition of energy each degree of freedom carries $\frac{1}{2}kT$. At room temperature a diatomic molecule can move in 3 directions and rotate about 2 axes; vibration is not excited, so the total is $\frac{5}{2}kT$. $\frac{3}{2}kT$ is only the translational part.

13. A metal block of mass 50 g at 300 °C is placed in a liquid of mass 160 g at 10 °C. The block is then quickly removed before thermal equilibrium is achieved and its temperature is found to have dropped to 120 °C. If the specific heat capacities of metal and liquid are 900 J kg^{-1} °C^{-1} and 2400 J kg^{-1} °C^{-1} respectively, what is the new temperature of the liquid?

A 120 °C B 63 °C C 53 °C D 31 °C

Answer: D

Heat lost by the metal = heat gained by the liquid:

$$(0.050)(900)(300 - 120) = (0.160)(2400)(\theta - 10)$$

$$8100 = 384(\theta - 10) \Longrightarrow \theta = 31 \text{ °C}$$

Explanation: The block is removed before thermal equilibrium, so the block and the liquid end at different temperatures. Use the temperature drop of the block (300 → 120 °C) to find the heat given out, then the rise in temperature of the liquid.

14. Which statement is true about the first law of thermodynamics?

A Energy is conserved in a system.
B The temperature of gas remains constant.
C No heat is allowed to enter or leave a system.
D The external work done is used to increase the internal energy of gas.

Answer: A

The first law, $Q = \Delta U + W$, is the principle of conservation of energy applied to a thermodynamic system.

Explanation: The heat supplied to a system goes into increasing its internal energy and doing work. B describes an isothermal process, C an adiabatic process, and D only one special case (adiabatic compression).

15. The human blood circulatory is considered a good model of forced convection because

A the heart pumps the blood and the blood maintains the body temperature
B the blood absorbs heat from the body and transfer the heat to all parts of the body
C the blood molecules vibrate faster and heat is transferred to other molecules to maintain the circulation
D the warm blood rises to upper parts of body due to the low density of the blood and the cool blood flows to the lower parts of the body

Answer: A

In forced convection the fluid is moved by an external device. The heart acts as a pump that drives the blood, and the circulating blood carries heat and keeps the body temperature steady.

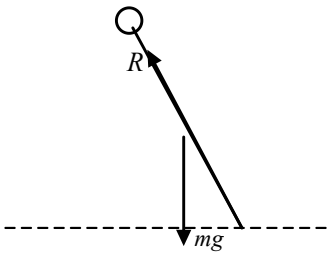
Explanation: Natural convection is driven by density differences (warm fluid rising), as described in D. The blood does not circulate because it is warm; it is pushed round by the heart, which is why it is a model of forced convection.

Section B [15 marks]

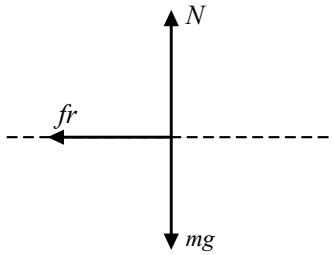
16. A cyclist is making a turn at a corner with a speed of 6.10 m s^{-1}.

(a) Sketch the forces that act on the cyclist and the bicycle. [2]

Answer



Resultant road force and weight



Components: N, friction fr and weight

Resultant road force R along the bicycle towards the centre (or normal reaction N and friction fr towards the centre) 1

Weight W / mg acting downwards 1

2

Explanation: Only two kinds of force act on the cyclist and bicycle: the weight and the force from the road. The road force can be drawn as one force along the leaning bicycle, or as its components: the normal reaction (up) and friction (horizontal, towards the centre of the turn). Friction provides the centripetal force.

(b) Determine the radius of the turning of the bicycle if the coefficient of kinetic friction between the tyre and the surface of the road is 0.34. [4]

Answer

$F_f = \mu mg$ 1

$F_c = \frac{mv^2}{r}$ 1

$F_f = F_c$ 1

$$\mu mg = \frac{mv^2}{r} \Longrightarrow r = \frac{(6.10)^2}{0.34(9.81)}$$

$r = 11.2$ m 1

[Accept 11.16 m or 11 m]

4

Explanation: Friction between the tyres and the road is the only horizontal force, so it provides the centripetal force. The mass cancels. A larger coefficient of friction allows a tighter turn (smaller radius) at the same speed.

(c) What will happen to the leaning angle of the bicycle if the speed of the bicycle is increased? [1]

Answer

The leaning angle increases 1

1

Total for question **7**

Explanation: A higher speed needs a larger centripetal force, so the horizontal component of the road force must be larger compared with the vertical component. The cyclist must lean further from the vertical: $\tan\theta = \frac{v^2}{rg}$.

17. (a) State the physical conditions for a real gas to behave as an ideal gas. Give reasons for your answers. [4]

Answer

Conditions: (very) high temperature 1

(very) low pressure 1

At high temperature the molecules move so fast that the potential energy of the molecules is negligible compared with their kinetic energy 1

At low pressure the molecules are very far apart, so the intermolecular forces are negligible 1

4

Explanation: An ideal gas assumes no intermolecular forces and negligible molecular volume. Both assumptions become good when the molecules are far apart (low pressure) and when their kinetic energy is much larger than any attraction between them (high temperature).

(b) The temperature of an ideal gas increases from 120 °C to 290 °C while the volume and the number of moles remain constant. Determine

(i) the ratio of the change in pressure, [2]

Answer

$pV = nRT$; with V and n constant, $p \propto T$ 1

$$\frac{p_2}{p_1} = \frac{290 + 273}{120 + 273}$$

$p_2 : p_1 = 1.43 : 1$ 1

2

Explanation: Always use the absolute temperature in gas laws. Using the Celsius values (290 ÷ 120) would wrongly give 2.42.

(ii) the ratio of the change in r.m.s. speed. [2]

Answer

$v_{\text{rms}} = \sqrt{\left(\frac{3kT}{m}\right)} \Longrightarrow \frac{v'_{\text{rms}}}{v_{\text{rms}}} = \sqrt{\left(\frac{T_2}{T_1}\right)} = \sqrt{\left(\frac{563}{393}\right)}$ 1

$v'_{\text{rms}} : v_{\text{rms}} = 1.20 : 1$ 1

[Alternative: $v_{rms} = \sqrt{\left(\frac{3pV}{m}\right)} \propto \sqrt{(p)}$, ratio = $\sqrt{(1.43)} = 1.20$]

2

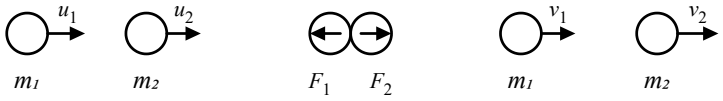
Total for question **8**

Explanation: The r.m.s. speed is proportional to the square root of the absolute temperature, so a temperature ratio of 1.43 gives a speed ratio of only $\sqrt{(1.43)} = 1.20$.

Section C [30 marks]

18. (a) By using Newton's second and third laws of motion, show the conservation of the linear momentum for two colliding objects. [3]

Answer



Before collision During collision After collision

From Newton's third law: $F_1 = -F_2$ (equal and opposite forces during the collision) 1

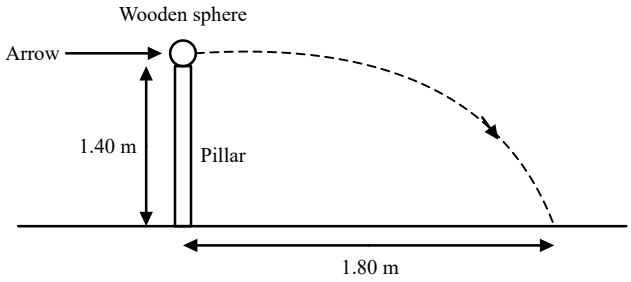
From Newton's second law: $\frac{m_1v_1 - m_1u_1}{t} = -\frac{m_2v_2 - m_2u_2}{t}$ 1

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ 1

3

Explanation: During the collision each object pushes on the other with equal and opposite forces for the same time, so the changes in momentum of the two objects are equal and opposite. The total momentum before the collision therefore equals the total momentum after.

(b) A wooden sphere of mass 128 g is at rest on top of a pillar of height 1.40 m as shown in the diagram below.



An arrow of mass 22.0 g is shot horizontally at the centre of the wooden sphere and stuck to it. The sphere and the arrow drop to the floor at a distance 1.80 m from the bottom of the pillar.

(i) Determine the time taken for the sphere and the arrow to reach the floor. [2]

Answer

$s = ut + \frac{1}{2}at^2$; $1.40 = 0 + \frac{1}{2}(9.81)t^2$ ($u_y = 0$) — 1

$t = 0.534$ s — 1

2

Explanation: Immediately after the collision the sphere and arrow move horizontally, so the initial vertical velocity is zero. The time of fall depends only on the height (1.40 m), not on the horizontal distance.

(ii) Calculate the speed of the arrow just before the arrow strikes the wooden sphere. [4]

Answer

Horizontal speed after impact: $v = \frac{s}{t} = \frac{1.80}{0.534} = 3.37$ m s^{-1} — 1

Total mass after impact: $m = (22.0 + 128)$ g — 1

Conservation of linear momentum: $m_1u = (m_1 + m_2)v$; $22.0u = (22.0 + 128)(3.37)$ — 1

$u = 23.0$ m s^{-1} — 1

4

Explanation: The horizontal velocity stays constant during the fall, so it can be found from the horizontal distance and the time from (i). The arrow sticks in the sphere (perfectly inelastic collision), so momentum is conserved but kinetic energy is not; the masses may stay in grams because they cancel.

(c) The first sphere of mass m_1 moves with velocity u and undergoes a head-on collision with the second sphere of mass m_2, which is initially at rest on a smooth horizontal surface. The collision is elastic.

(i) Define *elastic collision*. [1]

Answer

An elastic collision is one in which there is no loss in the total kinetic energy of the colliding objects (total kinetic energy is conserved). 1

1

Explanation: Momentum is conserved in all collisions, so the definition must mention kinetic energy and the word *total*: individual objects can gain or lose kinetic energy.

(ii) Deduce an expression for the velocity of the first sphere after collision in terms of m_1, m_2 and u. [3]

Answer

Momentum: $m_1u = m_1v_1 + m_2v_2 \Longrightarrow m_1(u - v_1) = m_2v_2 \ldots (1)$ 1

Kinetic energy: $\frac{1}{2}m_1u^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \Longrightarrow m_1(u^2 - v_1^2) = m_2v_2^2 \ldots (2)$ 1

(2) ÷ (1): $u + v_1 = v_2$; substitute in (1): $m_1u = m_1v_1 + m_2(u + v_1)$

$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u$ 1

[Alternative: use Newton's law of restitution for an elastic collision, $v_1 - v_2 = -(u_1 - u_2)$, with momentum conservation]

3

Explanation: An elastic collision needs two equations: conservation of momentum and conservation of kinetic energy. Dividing one by the other gives the useful result that the relative speed of approach equals the relative speed of separation.

(iii) Deduce the occurrence of the first sphere if both spheres have the same mass. [2]

Answer

For $m_1 = m_2$: $v_1 = 0$ 1

The first sphere stops (and the second sphere moves off with velocity u) 1

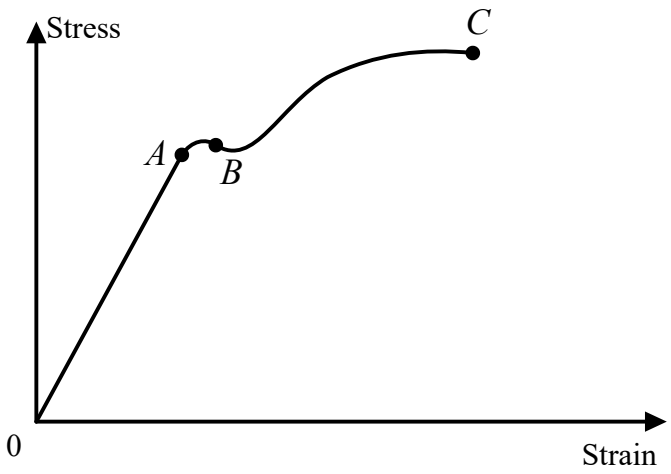
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Total for question **15**

Explanation: Equal masses in a head-on elastic collision exchange velocities. This is what happens with snooker balls and with Newton's cradle.

19. (a) By using a stress-strain graph, explain the proportional limit, elastic deformation and plastic deformation in a material. [9]

Answer



Correct graph	1
Label *OA*: proportional limit	1
Within the limit of proportionality the stress is directly proportional to the strain (obeys Hooke's law)	1
Label *OB*: elastic deformation	1
When the applied force is removed, the material returns to its original length	1
The net attractive force brings the molecules back to their equilibrium positions	1
Label *BC*: plastic deformation	1
When the applied force is removed, the material does not return to its original length	1
Dislocation: planes of molecules slide over each other	1
	9

Explanation: Describe each region in terms of what happens when the load is removed and what the molecules are doing. The proportional limit (A) is the end of the straight line; the elastic limit (B) is the last point from which the material can fully recover; beyond B the deformation is permanent.

(b) A hollow aluminium cylinder of length 40.0 cm with internal diameter 5.50 cm and external diameter 6.50 cm is compressed at each end with a force of 1.05 kN. Determine the

(i) stress on the cylinder, [3]

Answer

$A = \frac{1}{4}\pi(d_1^2 - d_2^2) = \frac{1}{4}\pi(0.065^2 - 0.055^2) = 9.42 \times 10^{-4}\ \text{m}^2$ 1

$\sigma = \frac{F}{A} = \frac{1.05 \times 10^3}{9.42 \times 10^{-4}}$ 1

$\sigma = 1.11 \times 10^6\ \text{N m}^{-2}$ 1

3

Explanation: The cylinder is hollow, so the force acts only on the ring-shaped (annular) cross-section: the area of the outer circle minus the area of the inner circle. Convert cm to m before squaring.

(ii) change in length due to the compression. [3]

Answer

$E = \frac{\text{stress}}{\text{strain}}$ 1

$\text{Strain} = \frac{\Delta l}{l}$ 1

$\Delta l = \frac{Fl}{AE} = \frac{(1.05 \times 10^3)(0.400)}{(9.42 \times 10^{-4})(7.00 \times 10^{10})} = 6.37 \times 10^{-6}\ \text{m}$ 1

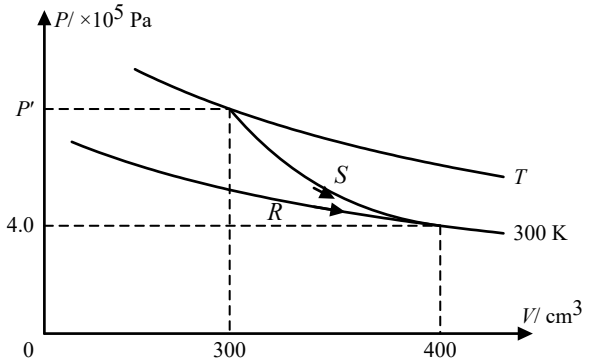
3

Total for question **15**

Explanation: Combine the definitions of Young's modulus, stress and strain. The change in length is tiny because aluminium is stiff; use the original length 0.400 m.

[Young's modulus of the aluminium is $7.00 \times 10^{10}\ \text{N m}^{-2}$.]

20. The variation of pressure, P, with volume, V, of one mole nitrogen gas for two processes, R and S, is shown in the graph below.



(a) Name R and S. [2]

Answer

R: isothermal (expansion) process	1
S: adiabatic (expansion) process	1
	2

Explanation: R follows the 300 K curve, so the temperature stays constant (isothermal). S is steeper and crosses from one isotherm to the other, so it is adiabatic.

(b) Describe the changes in the internal energy and the heat supplied in R and S. [7]

Answer

$Q = \Delta U + W$	1
In process R: constant temperature, $\Delta T = 0$	1
$\Delta U = 0$, no change in internal energy	1
$Q = W$: the heat supplied is used to do work in the expansion	1
In process S: $Q = 0$, no heat enters or leaves the system	1
The internal energy decreases	1
$\Delta U = -W$: the internal energy is used to do work in the expansion	1
	7

Explanation: The internal energy of an ideal gas depends only on its temperature. In the isothermal expansion all the heat supplied becomes work; in the adiabatic expansion no heat is supplied, so the work comes from the internal energy and the gas cools.

(c) Calculate the heat supplied in R. [2]

Answer

$W = nRT \ln \frac{V_2}{V_1} = (1)(8.31)(300) \ln \frac{400}{300}$	1
$Q = W = 717$ J	1
[Accept 717.2 J or 720 J]	
	2

Explanation: For an isothermal process $\Delta U = 0$, so the heat supplied equals the work done by the gas, $nRT \ln \frac{V_2}{V_1}$. The volumes may stay in cm^3 because only their ratio is needed.

(d) Calculate the work done in *S*. [4]

Answer

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$ 1

$\gamma = 1.4$ (diatomic) 1

$$(300)(400)^{0.4} = T(300)^{0.4} \Longrightarrow T = 336.6 \text{ K}$$

$W = \frac{nR}{\gamma - 1}(T - T_1) = \frac{8.31}{1.4 - 1}(336.6 - 300)$ 1

$W = 760$ J 1

Alternative (using pressure and volume)

$$P' = (4.0 \times 10^5)(\frac{400}{300})^{1.4} = 5.98 \times 10^5 \text{ Pa};\ W = \frac{1}{\gamma - 1}(P'V_1 - P_2V_2) = 48.8 \text{ J}$$

[Both methods are accepted. The data in the graph are not consistent with one mole at 300 K, so the two methods give different numbers.]

4

Total for question **15**

Explanation: For an adiabatic process use $TV^{\gamma-1}$ = constant (or pV^{γ} = constant) with $\gamma = 1.4$ for nitrogen, a diatomic gas. The work done by the gas equals the decrease in internal energy, $nC_V\Delta T = \frac{nR}{\gamma - 1}\Delta T$. Note: one mole at 300 K and 400 cm^3 would have a pressure of about 6×10^6 Pa, not 4.0×10^5 Pa, which is why the temperature method and the pressure method disagree.